

ABHISHEK V. KARVE

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EDUCATION

Stanford University

PhD in Applied Physics

Sept 2024 - Present

GPA: 3.96 / 4.00

- Research Assistant with Professor Jonathan Simon on millimeter wave to optical transduction using cold atoms.

University of Illinois Urbana-Champaign (UIUC)

Bachelor of Science in Engineering Physics with Minors in Mathematics and Computer Science

May 2024

GPA: 3.97 / 4.00

- Highest Honors
- Dean's List: Fall 2020, Spring 2021, Fall 2021, Spring 2022, Spring 2023, Fall 2023

RESEARCH EXPERIENCE

Simon Lab (simonlab.stanford.edu) (PI: Prof. Jon Simon)

Research Assistant

Stanford University

Sept 2024 to Present

- Building a next-generation quantum transducer using a Rydberg Rb85 atom ensemble in a hybrid millimeter-wave and optical cavity.
- Assisted the development of a novel 10 MHz Re-Imaging Phased Array Spatial Light Modulator (RIPA-SLM, [arxiv:2601.08906](https://arxiv.org/abs/2601.08906)) for high-speed optical control.

Covey Lab (coveylab.com) (PI: Prof. Jacob Covey)

Undergraduate Researcher | John A. Gardner Undergraduate Summer Research Award (2023)

University of Illinois Urbana-Champaign

January 2022 to May 2024

- Designed the vacuum chamber assembly for a new quantum computing experiment with modular transportable atom arrays.
- Implemented an AOM-based setup for focusing a beam through the AOM to achieve high switching speeds.
- Constructed a multi-level system simulation to look at qubit dynamics under Rydberg pulses.
- Built printed circuit boards to isolate digital and analog signals for experiment control for a new Ytterbium QIS experiment.
- Constructed circuit boards for amplification and noise-canceling of photo-diode output.

IBM / UIUC

Quantum UIUC/IBM Undergraduate Research Intern (PIs: Prof. Jacob Covey, Dr. Omar Shehab)

University of Illinois Urbana-Champaign

May 2022 to August 2022

- Built and tested a double-pass AOM setup to generate fast and frequency-tunable excitation pulses in Prof. Covey's Lab (UIUC).
- Researched dependence of required qubit circuit complexity on various graph properties while solving for MAXCUT using Quantum Approximate Optimization Algorithm (QAOA) with Dr. Shehab (IBM).

TECHNICAL SKILLS

- Proficient with optics alignment
- Proficient with through-hole and surface-mount soldering
- Proficient with Python, Mathematica
- Proficient with Autodesk Fusion 360, Eagle, Inventor
- Proficient in use of drill-press, end-mill
- Familiar with desktop CNC (Makera Carvera), lathe
- Familiar with C++, Java, MATLAB, Qiskit (qiskit.org)

PUBLICATIONS

Journal Articles

- X. Wei et al., A 10 Megahertz Spatial Light Modulator. *arXiv preprint* [arXiv:2601.08906](https://arxiv.org/abs/2601.08906) (2026), arxiv.org/abs/2601.08906
- Z. Li et al., A second-order optical Butterworth Fabry-Pérot filter. *Rev. Sci. Instrum.* **97**, 023103 (2026), doi.org/10.1063/5.0307543
- Z. Jia et al., An architecture for two-qubit encoding in neutral ytterbium-171 atoms. *npj Quantum Information* **10**, 106 (2024), doi.org/10.1038/s41534-024-00898-7

AWARDS

- Highest Honors** in Engineering Physics from the University of Illinois Urbana-Champaign. [2024]
- Dean's List** for Fall 2020, Spring 2021, Fall 2021, Spring 2022, Spring 2023, Fall 2023 Semesters. [2024]
- John A. Gardner Undergraduate Research Award** to support summer research under Dr. Jacob Covey. [2023]
- University of Illinois' **List of Teachers Ranked as Excellent** by Their Students for Spring 2023 based on student feedback. [2023]
- 1st Rank in India** for Computer Science (100 / 100) and Marathi(language) (99 / 100) in the Indian Certificate for Secondary Education. Awarded **1st Rank in School** for Science with a grade of 99 / 100 in the Indian Certificate for Secondary Education. [2018]

TALKS

Contributed

- Millimeter-wave Neutral-Atom Cavity QED for Quantum Interconnects. **Stanford QFARM Summer Talk Series**, (2026).
- Progress towards spin-squeezing in an ultracold neutral atom ensemble using a superconducting mmwave cavity. **57th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics**, (2026).
- Towards superconducting-cavity-induced spin squeezing in a neutral Rubidium-85 atom ensemble. **56th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics**, (2025).

CONFERENCE POSTERS

- A. V. Karve et al., Millimeter-wave Neutral-Atom Cavity QED for Quantum Interconnects. *57th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics*, (2026).
- Z. Li et al., Towards Quantum Internet Using Hybrid Quantum Technology. *56th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics*, (2025).
- W.K.C Sun et al., Towards a Rydberg neutral 171Yb atom array in a near-concentric dual-wavelength optical cavity. *55th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics*, (2024).

WORK EXPERIENCE

Department of Physics
Teaching Assistant | List of Teachers Ranked as Excellent by Their Students

- Taught a discussion section of 24 students for PHYS 211 - University Physics: Mechanics.
- Held weekly office hours for doubt resolution.
- Proctored midterm and final examinations.

University of Illinois Urbana Champaign
January 2023 to May 2023

ON-CAMPUS INVOLVEMENT

Physics Student Advisory Board (PhySAB)
Treasurer

Manage funds for PhySAB, place orders, and maintain a record of expenditures and funds. PhySAB is a UIUC Department of Physics advisory board that represents the undergraduate student body and advises the department administrators.

Department of Physics, UIUC
January 2023 to Present

Undergraduate Peer Mentoring
Peer Mentor

Act as a peer mentor to first-year students. Act as the first point of contact for informal queries about coursework and college life. Help create a 4-year college plan for the students to refer back to as a guide.

Department of Physics, UIUC
August 2022 to Present

Additional memberships: Society of Physics Students, University of Illinois Astronomical Society

VOLUNTEERING EXPERIENCE

NSF QUIC-TaQS Outreach to AVID Program
Volunteer Demo Assistant

Assisted Prof. Jacob Covey with demonstrations during a lecture at Franklin Middle School on information transmission and storage.

Urbana, IL
November 2023

Engineering Open House
Volunteer Exhibitor

Showed science demonstrations at the Society of Physics Students' booth. Engineering Open House is an event open to all where the students of the Grainger College of Engineering at UIUC show fascinating demonstrations and applications of science and engineering.

Urbana, IL
April 2022 and April 2023

Angel Xpress Foundation
Volunteer Teacher

- Volunteered as a teacher and taught Mathematics and English to middle school and high school underprivileged students.
- Developed teaching techniques to increase the concentration and enthusiasm of the students.

Mumbai, India
October 2018 to September 2019

RELEVANT GRADUATE COURSEWORK

General Physics Coursework

- **PHYSICS 212: Statistical Mechanics** Ising model, qualitative properties, diagnostics, correlation functions, high-temperature and low-temperature expansions; one-dimensional Ising model; Kramers-Wannier duality; Z2 lattice gauge theory; Coarse-graining and mean field theory; Consequences of mean field theory; gaussian integrals and correlation functions; spontaneous symmetry breaking.
- **PHYSICS 220: Classical Electrodynamics** Special relativity, Lorentz transformations, four vectors and tensors; Lagrangian formulation, charges, gauge invariance, field tensor, covariant equations, four-current, continuity equation; Noether's theorem, Poynting's theorem, stress-energy tensor; conductors and dielectrics, magnetic media; Plane and monochromatic waves, spectral resolution, dispersion relations, wave guides.
- **PHYSICS 230: Graduate Quantum Mechanics** Hilbert spaces, Dirac's notation; periodic structure; symmetry operations, gauge transformation; path integral formulation; measurement theory; angular momenta, central potential problems.
- **PHYSICS 216: Back of the Envelope Physics** Order of magnitude approaches to estimating physical effects; scaling, dimensional analysis.

Atomic, Molecular, Optical Physics Coursework

- **APPPHYS 203: Atoms, Fields and Photons** States, pictures, harmonic oscillators; 2-level systems; Bloch sphere; spontaneous emission; Fermi's rule; Open systems; fine structure, Lamb shift, hyperfine structure, Zeeman shift; scattering; optical Bloch equation; optical pumping; laser cooling, MOT;
- **APPPHYS 282: Ultracold Quantum Physics** Quantum optics and atoms in the ultracold setting; quantum gases and photons; laser cooling and trapping, optical lattices; cavity QED; BEC and quantum degenerate Fermi gases, ultracold collisions, quantum phase transitions in quantum gases and lattices.
- **EE 236A: Modern Optics** Geometrical optics; lens analysis and design, aberrations, optical instruments; ray matrices; wave nature of light; polarization, plane waves at interfaces and in media ; diffraction; Fourier Optics; Gaussian beams.
- **EE 340: Quantum Photonics** Generation and manipulation of quantum light on a chip; coherent Fock, squeezed, entangled states of light; cavity quantum electrodynamics, strong and weak-coupling regime; solid state light-matter entanglement; photon statistics and photon indistinguishability; quantum state tomography.

Quantum Computing Coursework

- **EE 224: Quantum Control and Engineering** Quantum control, dynamics, and information processing; qubits and operators; modeling and numerical analysis of open quantum systems; quantum control protocols; average Hamiltonian theory; dynamical decoupling; quantum device benchmarking; different quantum platforms and their applications.

Mathematical Coursework

- **EE 261: The Fourier Transform and Its Applications** Fourier transform as a tool for solving physical problems; Fourier series; Fourier transform of continuous and discrete signals and its properties; Dirac delta, distributions, generalized transforms; convolutions and correlations and applications; probability distributions, sampling theory, filters, and analysis of linear systems; discrete Fourier transform and the FFT algorithm; multidimensional Fourier transform; applications to optics, crystallography.

RELEVANT UNDERGRADUATE COURSEWORK

Experimental Coursework

- **PHYS 403: Modern Experimental Physics** Modern experimental techniques; quantum optics: optical alignment and design; fluorescence spectroscopy; signal analysis, nitrogen cryostat; atomic force microscopy; second sound in helium: helium cryostat, lock-in amplifiers, data analysis. Oral presentations of work done; detailed laboratory reports.
- **PHYS 498EBP: Experimental Biophysics** Aseptic techniques; micro-pipette techniques; polymerase chain reaction procedure; microscopy using nitrogen-vacancy centers in nanodiamonds; studying E. coli and onion cells in optical tweezers; Luria Delbruck experiment; super-resolution microscopy.

Theory Coursework

- **PHYS 325: Classical Mechanics 1** Newtonian Kinematics and Dynamics; three-dimensional motion, variable mass, and conservation laws; damped and periodically driven oscillations; gravitational potential of extended objects and motion in rotating frames of reference; Lagrangian and Hamiltonian mechanics.
- **PHYS 326: Classical Mechanics 2** Coupled oscillators, chaos dynamics, rotational motion, continuous media, and elementary fluid dynamics.
- **PHYS 435: Electromagnetic Fields 1** Static electric and magnetic fields; field interactions with electric charge and current; field transformation properties; derivation of Maxwell's equations; fields in linear media.
- **PHYS 436: Electromagnetic Fields 2** Time-dependent fields; electromagnetic induction; electromagnetic wave propagation in various media and structures; electromagnetic radiation from charge and current distributions; relativistic covariance of Maxwell's equations.
- **PHYS 486: Quantum Mechanics 1** The Schrodinger equation and it's properties; wave functions; single particles in 1D in square well, quantum harmonic, and particle-in-a-box potentials; Hilbert spaces, Dirac notation, and operators; angular momentum, Hydrogen atom; spin; identical particles; brief introduction to quantum information.
- **PHYS 487: Quantum Mechanics 2** Hamiltonian symmetries; perturbation theory; spin-orbit coupling; fine structure of hydrogen; Zeeman effect; variational principle; emission and absorption of radiation; adiabatic theorem; scattering theory; free electron gas; electrons in solids.
- **PHYS 427: Thermal & Statistical Physics** Micro-canonical ensembles; grand canonical ensemble and partition functions; ideal gas formulations; Debye model; quantum statistics; systems with interactions; van der Waals model; Weiss model of ferromagnetism; Landau theory; kinetic theory: conduction, radiation and propagation.

Mathematical Coursework

- **MATH 347: Fundamental Mathematics** Techniques of proof writing; mathematical induction; binomial coefficients; rational and irrational numbers; least upper bound axiom for real numbers; convergence of sequences and series.
- **MATH 416: Intro to Linear Algebra** Determinants; vector spaces over fields; linear transformations; inner product spaces; eigenvectors and eigenvalues; Hermitian matrices.
- **MATH 442: Intro to Partial Differential Equations** Wave, diffusion, and potential (Laplace) equations; fundamental solutions and transform methods for problems on the line; separation of variables using orthogonal series boundary value problems; convergence of Fourier series.
- **MATH 417: Intro to Abstract Algebra** Fundamental theorem of arithmetic; congruences; permutations; groups and subgroups; homomorphisms; group actions with applications; polynomials; rings, subrings, and ideals; integral domains and fields; roots of polynomials; maximal ideals; construction of fields.
- **STAT 400: Statistics and Probability 1** Mathematical statistics; calculus of probability; random variables; expectation, distribution functions; central limit theorem, point estimation; confidence intervals; hypothesis testing.

Computational Coursework

- **CS 225: Data Structures** Data abstractions: elementary data structures (lists, stacks, queues, and trees) and implementation using C++; solutions to computational problems like search on graphs and trees; elementary analysis of algorithms.
- **CS 357: Numerical Methods** Floating-point computation; systems of linear equations; approximation of functions and integrals; single nonlinear equation; numerical solution of ordinary differential equations; various applications in science and engineering; programming exercises and use of high-quality mathematical library routines.